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Putnam 2005 A1

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Show that every positive integer is a sum of one or more numbers of the form $2^r 3^s$, where r and s are nonnegative integers and no summand divides another. (For example, $23 = 9 + 8 + 6$.)

Thoughts I

1. The problem ask for every positive integer. So I should give a notation for it.
2. Let n be a positive integer.
3. We roughly want to show $n = \sum_{r,s} 2^r 3^s$.
4. (3) of course is not rigorous; r and s here are subject to the constraint given in the problem.
5. I recall what is summand here: In $23 = 9 + 8 + 6$, the summands are 9, 8, 6.

Thoughts II

6. No summand divides another: I try to understand this using examples.
- 6.1 If I have 6, then I know other summand cannot be a multiple of 6 (e.g. 12, 18, ...). I also, I cannot have divisors of 6 (e.g., 1, 2, 3).
- 6.2 Generally, If I have $2^r 3^s$ ($r, s \geq 0$), then other summand cannot be of the form:
- ▶ $2^a 3^b$ where $0 \leq a \leq r$ and $0 \leq b \leq s$
 - ▶ $2^c 3^d$ where $c \geq r$ and $d \geq s$.
- 6.3 From (6.2), the other summand must be of the form $2^a 3^b$ where (a, b) satisfies $0 \leq a \leq r$ and $b > s$; or $a > r$ and $0 \leq b \leq s$ (one exponent is larger while the other one is smaller).
- 6.4 I observe that if 1 is a summand of n , then this implies $n = 1$. This is because if n contains another summands, they will be divisible by 1, a contradiction.

Thoughts III

7. Let's try small cases to see any pattern here:

7.1 $n = 1$: $n = 2^0 3^0$.

7.2 $n = 2$: $n = 2^1 3^0$.

7.3 $n = 3$: $n = 2^0 3^1$.

7.4 $n = 4$: $n = 2^2 3^0$.

7.5 $n = 5$: $n = 2 + 3 = 2^1 3^0 + 2^0 3^1$.

7.6 $n = 6$: $n = 2^1 3^1$.

7.7 $n = 7$: I cannot immediately get the answer because my calculation is bad. I see that I cannot take $2 + 5$ because $2^1 3^0$ appears twice. $1 + 6$ also cannot. Oh found it
 $7 = 4 + 3 = 2^2 3^0 + 2^0 3^1$.

7.8 $n = 8$ and 9 : $8 = 2^3 3^0$ and $9 = 2^0 3^2$.

7.9 $n = 10$: $10 = 4 + 6 = 2^2 3^0 + 2^1 3^1$.

7.10 $n = 11$: $11 = 5 + 6$ cannot. $11 = 9 + 2$ works.

Thoughts IV

8. I introduce an informal notation to see it clearer
 $n = (a, b) + (c, d)$ means $n = 2^a 3^b + 2^c 3^d$:

8.1 $1 = (1, 0)$

8.2 $2 = (1, 0)$

8.3 $3 = (0, 1)$

8.4 $4 = (2, 0)$

8.5 $5 = (1, 0) + (0, 1)$

8.6 $6 = (1, 1)$

8.7 $7 = (2, 0) + (0, 1)$

8.8 $8 = (3, 0)$

8.9 $9 = (0, 2)$

8.10 $10 = (2, 0) + (1, 1)$

8.11 $11 = (0, 3) + (1, 0)$

Thoughts V

9. (★) I feel that odd number is harder to break down. This motivates to start a new work: Split into even and odd.
10. Analyze the patterns for $n = 2, 4, 6, 8, 10, \dots$. It seems like I can built n based on smaller integers.
11. Observe that $4 = 2 \cdot 2$ and $8 = 2 \cdot 4$, which explains why there is a pattern
$$\begin{array}{ccccc} 2 & \longrightarrow & 4 & \longrightarrow & 8 \\ (1,0) & & (2,0) & & (3,0) \end{array}$$
.
12. Also observe that
$$\begin{array}{ccc} 5 & \longrightarrow & 10 \\ (1,0)+(0,1) & & (2,0)+(1,1) \end{array}$$
.
13. There is a connection between $n/2$ and n : The exponent of 2 in each summand will increase by 1.
14. More formal claim: If $n/2 = \sum_{r,s} 2^r 3^s$, then $n = \sum_{r,s} 2^{r+1} 3^s$.
15. Multiplying by 2 does not affect the divisibility here; what I mean is if a does not divide b , then $2a$ does not divide $2b$.
16. (★) Combine (14) and (15). I see that the even case is done; and a rigorous argument can be written using strong induction if odd case can be worked out similarly ($n/2$ can be odd, not we cannot use induction on even n).

Thoughts VI

17. Every case $n = 2^r 3^s$ is considered done.
18. By (16), we have finished the even cases and the case $n = 3^s$ is also obvious. So for the odd case, we left with $n \neq 2^r 3^s$ (at least two summands).
19. $n - 3^a$ is always even.
20. If $n \neq 3^a$ has at least two summands, then I can use induction hypothesis to show $n - 3^a = \sum_{r,s} 2^r 3^s$. However, this idea seems to have a flaw as 3^a may appear in $\sum_{r,s} 2^r 3^s$. Need improvement to proceed.
21. Take the largest exponent a of 3 so that $3^a < n < 3^{a+1}$. Use (20): $n - 3^a = \sum_{r,s} 2^r 3^s$. We still cannot ensure whether divisors of 3^a appears as a summand. It will be good if there is no summand of the form 3^s , can we force the summands to have 2?

Thoughts VII

- 22. (19) tells us that $\frac{n-3^a}{2}$ is also a positive integer.
- 23. So $\frac{n-3^a}{2} = \sum_{r,s} 2^r 3^s$.
- 24. (★) So $n - 3^a = \sum_{r,s} 2^{r+1} 3^s$, this resolve the issue in (21).
- 25. If 3^a does not divide any other summands, then we are done.
- 26. (★) (25) is true if we impose (21) and $n \neq 2^r 3^s$ from (18).
- 27. To see (26), note that $3^a < n < 3^{a+1}$. If 3^a divides a summand in (24), a summand must be of the form $2 \cdot 3^a$. But $n - 3^a = \sum_{r,s} 2^r 3^s \geq 2 \cdot 3^a \implies n \geq 2 \cdot 3^a + 3^a = 3^{a+1}$, a contradiction.
- 28. The odd case is done. We can start writing the solution.

Solution I

We want to show that for every positive integer n ,

$$n = \sum_{i=1}^k 2^{r_i} 3^{s_i}$$

for where $k \geq 1$, and r_i, s_i ($i = 1, \dots, k$) are nonnegative integers such that $2^{r_i} 3^{s_i} \nmid 2^{r_j} 3^{s_j}$ for all $j \neq i$.

We prove this by strong induction on n .

Base step: $n = 1 = 2^0 3^0$. So the result is true.

Inductive step: Let $n > 1$ be a positive integer. Suppose that the result holds for every positive integer $k < n$.

Solution II

Case 1: n is even.

Then $n = 2m$ for some integer $1 \leq m < n$.

By inductive hypothesis,

$$m = \sum_{i=1}^k 2^{r_i} 3^{s_i}$$

where no summand divides another.

So $n = \sum_{i=1}^k 2^{r_i+1} 3^{s_i}$. Note that no summand on RHS divides another. Hence Case 1 is done.

Case 2: n is odd.

Case 2.1: $n = 3^a$ for some $a \geq 1$. There is nothing to do.

Case 2.2: n is not a power of 3.

We can find an integer $a \geq 0$ such that $3^a < n < 3^{a+1}$.

Since n and 3^a are odd, we see that $n - 3^a$ is a positive even integer.

Solution III

Hence $\frac{n-3^a}{2}$ is a positive integer less than n .

By induction hypothesis, $\frac{n-3^a}{2} = \sum_{i=1}^k 2^{r_i+1}3^{s_i}$ where no summand divides another.

So $n = 3^a + \sum_{i=1}^k 2^{r_i+1}3^{s_i}$.

We show that 3^a does not divide and is not divided by $2^{r_i+1}3^{s_i}$ ($i = 1, \dots, k$).

Clearly $2^{r_i+1}3^{s_i} \nmid 3^a$.

Suppose for the sake of contradiction that $3^a \mid 2^{r_i+1}3^{s_i}$ for some i .

Then $a \leq s_i$.

Hence $2^{r_i+1}3^{s_i} \geq 2 \cdot 3^a$.

Solution IV

Since $2^{r_i+1}3^{s_i}$ is one of the summands in the sum $\sum_{i=1}^k 2^{r_i+1}3^{s_i}$, we see that

$$n = 3^a + \sum_{i=1}^k 2^{r_i+1}3^{s_i} \geq 3^a + 2 \cdot 3^a = 3^{a+1},$$

this contradicts $n < 3^{a+1}$.

Hence Case 2 is done.

By strong induction, the result is true for all positive integer n .

Remarks I

1. Choosing effective notations matter. One should always write simplify the objects by giving more convenient notations. In this case, I will not identify the pattern quickly if I do not write $2^a 3^b$ as (a, b) , although It does not show up in solution.
2. Parity (even and odd) often splits the problem into doable cases.
3. Doubling trick does solve the even case. In many problems, we can always attack the problem by relating n and $n/2$, algebraically or combinatorically.

Remarks II

4. Do not give up everything when you find a flaw in your argument. Since it might be a potential solution, look for extra argument that is enough to fix or avoid the flaw.
5. Always consider extreme values. In many problems (analysis, combinatorics), we will start attacking them by looking at the behavior of the largest / smallest elements (in some sense). Extreme points usually has extra constraint compared to the other.